

Agility Performance and Change-of-Direction Ability

In both field and court sports, there is a substantial number of preplanned COD movements as well as changes in direction in response to the ball, the game, teammates, or opponents. In sports such as baseball, softball, American football, and basketball, athletes decide before commencement of movement on the path they will run. Such patterns often including a quick or sudden COD—for example, exploding into a sprint out of a shuffle off a base or running a route before receipt of a ball. However, many sports also include rapid COD in response to teammates, opponents, or tactical situations (i.e., external stimuli). Such responses can occur in an offensive or a defensive scenario, and the physical movement and perceptual-cognitive aspects of such performances impose different demands on the athlete. Therefore, one should thoroughly understand the factors associated with agility performance from a physical standpoint (COD ability) and in combination with the perceptual-cognitive aspects (agility).

Athletes improve COD ability through development of several physical factors and technical skills during a variety of speeds and modes of movement. The development of agility also requires improving perceptual-cognitive abilities in relation to the demands of the sport.

Factors Affecting Change of Direction and Perceptual-Cognitive Ability

It is becoming increasingly clear that the ground contact time and GRF during the plant phase ([figure 20.4](#) on [page 643](#)) of a movement provide valuable insight into the physical factors that affect COD performance, while perceptual-cognitive factors must

also be considered regarding agility. COD and agility movements performed at shallow cutting angles (less than 75 degrees) and associated with shorter ground contact times (less than 250 ms) (94) benefit from training similar to speed training with regard to the physical demands but still require additional perceptual-cognitive training. On the other hand, when a COD involves a more aggressive cutting angle (equal to or greater than 75 degrees), the length of ground contact time can often exceed 250 milliseconds (8, 48, 64, 36) due to the greater braking requirements. Therefore, one should consider increasing emphasis on eccentric strength for braking and maximal strength alongside the concentric impulsiveness required for propulsion during the reacceleration. Additionally, due to the eccentric–concentric coupling during the plant phase of directional changes, reactive strength development is also a key training priority (36). [Figure 20.11](#) provides an example of how various requirements (angle of COD or perceptual-cognitive requirement of agility) may affect the GRF and ground contact times of athletes. Of particular interest is the effect of various methods of changing direction (for the same angle of direction change) on GRF and ground contact times. The effect of method of changing direction can be seen in [figure 20.11](#): The 5-0-5 was performed either with a single-leg COD (outside leg turn) or with a jump turn that can be performed in a testing environment but is not commonly performed on the field of play. Nevertheless, this demonstrates that the technique chosen dictates the demands; therefore, if trying to elicit a specific adaptation from an athlete, one should provide specific instructions.

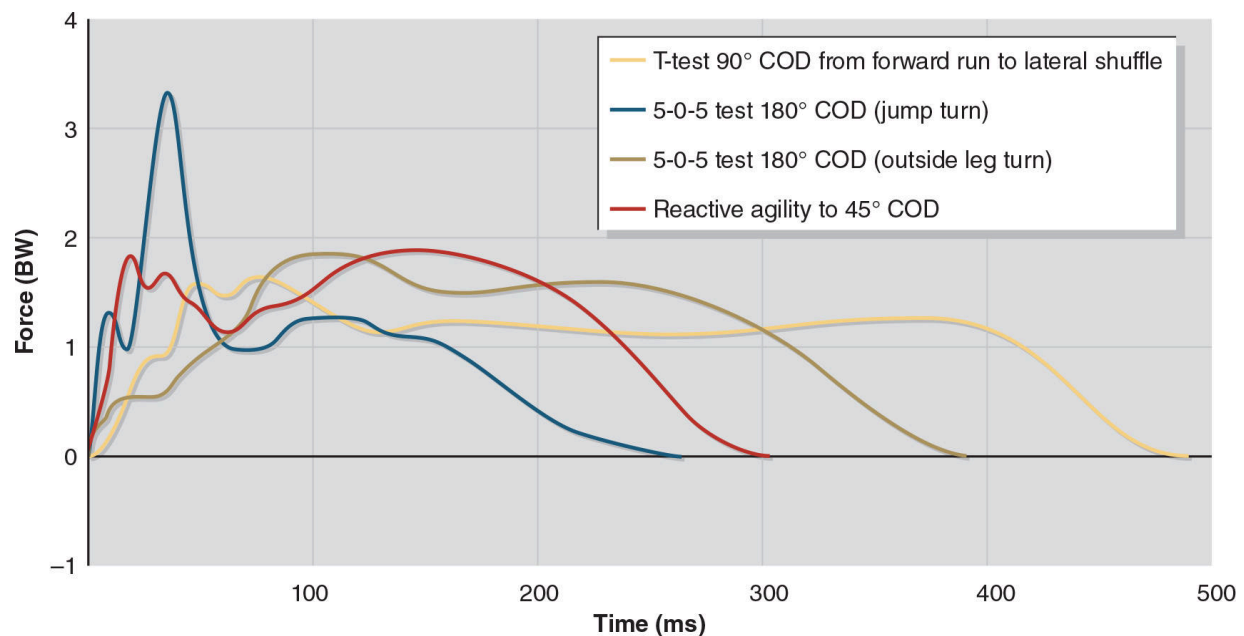


FIGURE 20.11 Comparison of ground reaction force and contact times during various change-of- direction and agility maneuvers.

The test used for assessment of agility performance has an impact on perceived agility ability of the athlete. A plethora of tests are used to assess “agility,” but tests without a reactive aspect should be considered COD tests, and a test that includes a reactive stimulus is now by definition a test of agility for most sports. Further, the requirements of the test may result in assessment of factors other than COD ability or perceptual-cognitive requirements. Therefore, one must be informed about and critical regarding a test used to assess “agility” or “change-of-direction ability.” For example, if the Illinois agility test, often performed with soccer athletes (91), was used after preseason training, improvements in performance may be due to improved metabolic capacity and not necessarily improved COD ability. This is attributed to the length of the Illinois test, which imposes a higher metabolic demand than shorter tests and thus is more representative of **maneuverability**.

However, it should be noted that even similar-length tests such as the traditional 5-0-5 test, the modified 5-0-5 test (the 5-0-5 without the initial 10 m run-up), and the L-run used on the same athletes

have only a moderate relationship (41), indicating they are not all assessing the same type of COD ability or the same underpinning physical requirements. This becomes more apparent when one realizes that some tests require a rapid COD while others require multiple COD. Tests with multiple COD often require more bending around an object or cone while maintaining as much velocity as possible, in comparison to a sharp COD that demands a rapid deceleration (e.g., 5-0-5 test). Therefore, tests such as the Illinois agility or L-run that require a bend to maintain velocity instead of an aggressive COD could be considered tests of maneuverability (69). With this said, [table 20.3](#) gives examples of COD and agility tests and drills that address different physical requirements within the realm of “agility.” One should consider the aspect most important to the given sport when determining COD and agility abilities and consider testing and monitoring various types of COD abilities and perceptual-cognitive abilities required during true agility tests. Furthermore, tests such as the traditional 5-0-5 may be influenced by an athlete’s linear running speed (70, 81); thus, to isolate COD ability, the COD deficit (5-0-5 test time—10 m linear sprint time) could be determined (70, 71). Practitioners may also want to consider an athlete’s sprint momentum (average velocity during a linear sprint test or 10 m approach of the traditional 5-0-5 test × athlete’s mass) to provide an indication of the braking demand experienced by athletes during the 5-0-5 test to give a more complete view of the athlete’s COD ability.

TABLE 20.3 Aspects of Agility Tested or Trained in Various Drills and Tests

Change-of-direction speed	Maneuverability	Perceptual-cognitive ability	Metabolic requirement ^a (average length of test or drill)
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	Change-of-direction speed	Maneuverability	Perceptual-cognitive ability	Metabolic requirement ^a (average length of test or drill)
TESTS				
Reactive agility tests	✓		✓	<3 s
5-0-5	✓			<3 s
5-10-5	✓			<5 s
T-test	✓	✓		<12 s ^a
Illinois agility		✓		<12 s ^a
L-run		✓		<6 s
DRILLS				
Z-drill	✓	✓		<5 s ^b
Y-shaped agility drill (generic stimulus)	✓			<5 s ^b
Y-shaped agility drill (specific stimulus)	✓		✓	<5 s ^b

^aThe length of the test has a large influence on the metabolic requirements and whether changes in test performance can become a function of enhanced conditioning instead of directly related to improvements in COD or agility ability.

^bDuration dependent on distances used within the design of the drill. However, it is advisable to keep the duration of the drill short to emphasize technical execution within the drill.

Data from Gabbett et al. (2008); Nimphius et al. (2012); Pauole et al. (2000); Serpell et al. (2011); Sierer et al. (2008); Vescovi and McGuigan (2008); Vescovi et al. (2011); Young and Farrow (2013).

Change-of-Direction Ability

As stated earlier, one must understand that an athlete's measured COD ability may vary depending on the demand imposed by the

chosen COD test. Therefore, it may be beneficial to choose different tests that measure this ability under a high-velocity braking requirement; with multiple COD; or, as will be discussed, in reaction to a scenario, opponent, or stimulus. Another consideration regarding COD ability is the orientation of the body leading into the deceleration, through the transition phase as the athlete comes to a stop, and then the positioning as he or she subsequently reaccelerates out of the COD. Therefore, the positioning of the trunk influences the performance of the COD. As such, it is a combination of the ability to decelerate (high braking impulse), reorient the body to face or partially face the direction of intended travel, and then reaccelerate (high propulsive impulse) over short contact times that truly determines COD ability. Additionally, the ability to perform the COD requires the athlete to have the physical capacity to produce high and rapid magnitudes of force, tolerate the high loads associated with braking, and adopt favorable postures that are associated with braking and propulsion (22, 35).

Considering the requirements for rapidly changing momentum, increased muscle mass in combination with decreased body fat (i.e., improving functional mass) has been regarded as a predictor of COD performance when assessed by the T-test in elite basketball players (23). Other anthropometrics and their relationship to COD ability are not discussed here, because aspects such as height and limb length are not modifiable by the strength and conditioning professional. Instead, body center of mass height is related to improved lateral COD performance (89), and this can be actively targeted during technique modification training (22, 35).

Overall, COD ability among a variety of movement patterns (various angles of direction change) has been shown to improve with increased hip extension velocity (rapid application of force by the hip extensors), low center of mass height, wide lateral foot plants for perpendicular force application, increased braking impulse and propulsive impulse, increased knee flexion entering the COD, minimized trunk angular displacement entering the COD

(deceleration phase), and increased lateral trunk tilt (during 180-degree changes) (22, 35, 80, 89, 94, 95, 97). Because the ability to decelerate, hold body position, and then reaccelerate is important in COD tasks as described by the kinetic and kinematic requirements of faster COD and agility performances, it is clear that a well-rounded approach to strength development involving dynamic, isometric, and in particular eccentric strength capacities is needed for the development of better COD performances (51, 97), with training methods that span the entire load–velocity and force–velocity curve (42) across multiple lower-limb and trunk segments (35). In contrast to sprint development, it is recommended that athletes undergo training that allows for the neuromuscular system to effectively adapt to the rate of loading required during the braking phase, with the understanding that the neuromuscular requirements for braking capacity must be specifically trained using low- and high-velocity eccentric muscle actions (38) such as those during drop landings, landing from a loaded jump, or the catch phase of a power clean or power snatch, in addition to supramaximal eccentrics that particularly load the knee extensors. All of these place different eccentric loading demands on the hips, knees, and ankles (67).

The biomechanical demands of directional changes are influenced by approach speed and COD angle; thus, practitioners should manipulate these two variables cautiously when designing COD and agility activities.

Perceptual-Cognitive Ability

When the physical ability to change direction rapidly is present, one can focus on the components of perceptual-cognitive ability that must be developed to bring together the ability to perform both physically and mentally on the field of play. There are several components of perceptual-cognitive ability: visual scanning, anticipation, pattern recognition, knowledge of the situation, decision-making time and accuracy, and reaction time (86, 88, 94,

110, 111). Many of these aspects of development are sport specific, and a comprehensive discussion is beyond the scope of this text. However, general drills to help improve these skills are discussed in the section on methods of developing agility.

Technical Guidelines and Coaching

In comparison to sprinting, COD and agility have a large degree of freedom due to the multitude of movements that occur during a COD. Further, agility performance as restricted or determined by opponents or other tactical restraints and scenarios cannot be trained through the use of a single technique, with evidence to suggest we should be creating within- and between-skill variability and a flexible coordinative approach (35). Nevertheless, the following are some technical guidelines and coaching suggestions.

Visual Focus

- When changing direction in response to an opponent (either offensive or defensive), the athlete should visually scan the environment and opponent while focusing on the shoulders, trunk, and hip.
- Following the anticipation of the event, unless deception is intended, the athlete should quickly redirect attention to a new area to help lead the transition of the body.

Body Position During Braking and Reacceleration

- Control the trunk leading into the deceleration (decrease large amounts of trunk motion) (80). When decelerating, posterior trunk lean may assist in orienting the force vector posteriorly.
- Through the stance phase, reorient the trunk and hips toward the direction of intended travel to allow for a more effective reacceleration (22) and to reduce the redirection demands for a more effective position to produce force (35).
- Just as with acceleration mechanics, body lean is paramount in allowing proper force application through the ground with

strong alignment of the ankle, knee, and hip to the trunk and shoulders.

- Enter and exit changes in direction with a lower center of mass to increase stability and to enable a manipulation of the base of support relative to the center of mass (89).

Leg Action

- Ensure that the athlete can effectively dissipate or tolerate the eccentric braking loads through an effective range of motion at the hip, knee, and ankle (i.e., co-triple flexion) and avoid a stiff-legged braking style (92, 94).
- Emphasize a firm base of support for effective push-off and manipulation of base of support relative to center of mass (i.e., wide lateral foot plant when cutting, large anterior foot placement when turning) while pushing the ground away in order to enhance performance (i.e., rapid simultaneous triple extension of ankle, knee, and hip), especially while learning in closed drills. External focus of attention—through instructions to concentrate on the ground instead of a body part—has been shown to improve COD performance (76).

Arm Action

- Powerful arm actions should be used to facilitate leg drive.
- Ensure that the action of the arms is not counterproductive (i.e., does not cause a decrease in speed or efficiency), particularly during transitioning between difficult COD (e.g., from a backpedal to a sprint), while keeping the arms close to the body to reduce the moment of inertia and to increase the total body rotation velocities when changing direction (35).

Transitional Movements

Expressions of agility can be divided into three phases (21): a **transition** whereby the athlete is in a ready position to read and subsequently react to a situation that arises; an **initiation**, a moment when an athlete initiates or changes movement (e.g., a

COD); this leads to the **actualization** phase that involves a sport skill or sprint to a position. For example, a corner back, when tracking a wide receiver, may perform a backpedal (transition), before performing a directional change (initiation) into a sprint and then intercepting an underthrown pass (actualization). While the success of this play or other expressions of agility manifest in the actualization phase, the preceding phases are just as important and require good technical execution. **Backpedaling** (rearward locomotion with periods of double limb support and one foot in contact with the ground between contralateral foot strikes) or **backward running** (rearward locomotion involving single-leg support throughout foot-ground contact and both feet simultaneously in the air between foot strikes) (101) may be performed to displace an athlete rearward while maintaining a forward visual gaze of an opponent or ball, such as in basketball, soccer, rugby, and football. The frequency and importance of differing transitional movements may be sport dependent, and more research is required in specific sports to develop this understanding (12, 21). Nevertheless, practitioners should develop their athletes' ability to perform a variety of transitional movements (e.g., lateral shuffle, backpedal, crossover running) and to perform sprints from these transitional movements to develop athletes' movement solutions and all-around physical literacy.

[Figure 20.12](#) provides a technique checklist for common transitional movements: the backpedal and side-shuffle. The backpedal is an essential locomotive skill for a defensive back in American football. The following drill progression could be used to develop an athlete's backpedaling technique before integrating the skill with other actions or agility drills:

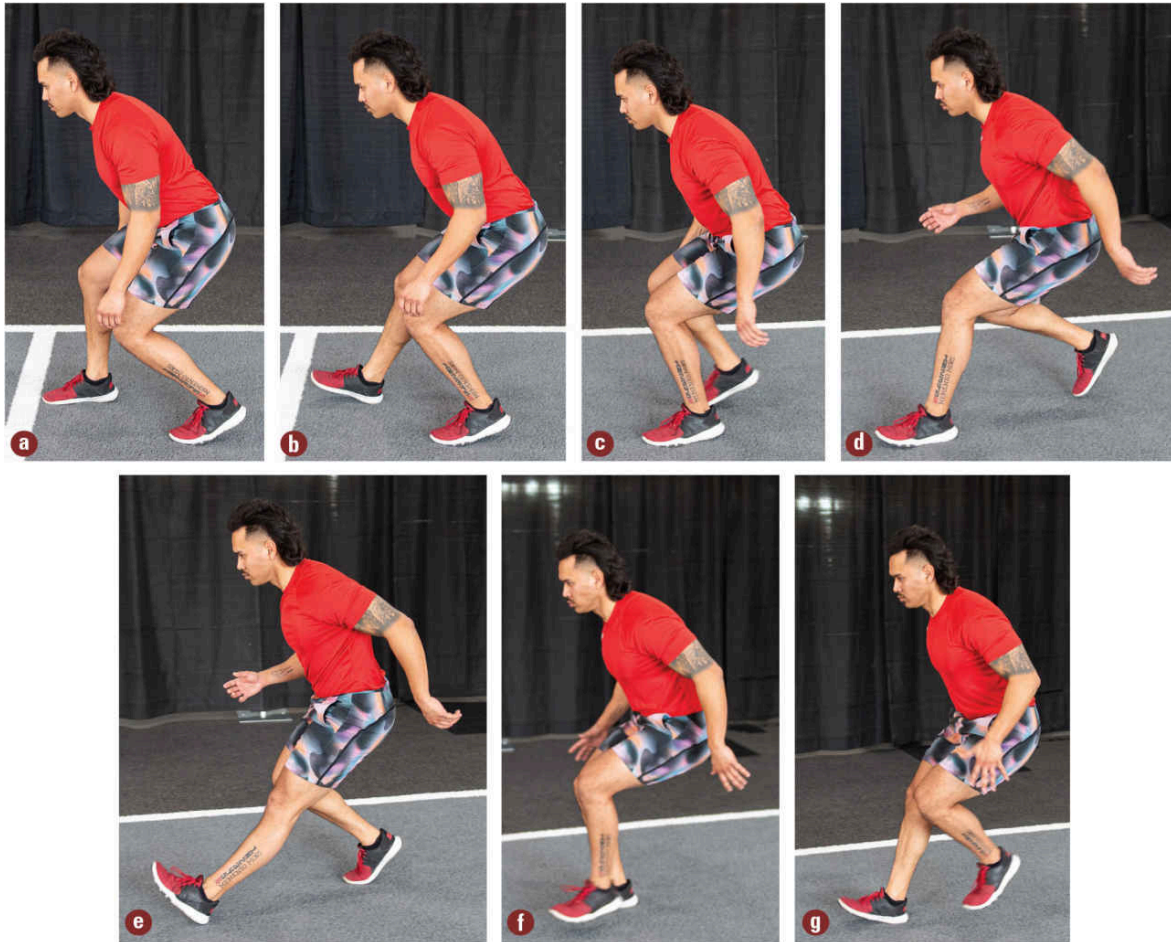
- Backpedaling for 20 yards (18 m) along a marked line on the football field to establish correct technique
- Backpedaling for 20 yards (18 m), changing the pace of the drill via an audio command (e.g., "slow" or "fast"), to develop

the athlete's ability to adapt backpedaling speed in line with tracking an opponent

- Backpedaling and then either on command or when reaching a checkmark changing direction (either forward, backward, or lateral to either side)
- Backpedaling while tracking an opponent who is initially running forward and then changing direction to sprint once the opponent initiates a COD or sprint

A similar progression could be used for other transitional movements such as side-shuffling in court-based sports such as basketball and tennis.

Figure 20.12 Technique checklist for common transitional movements: backpedal and side-shuffle.



Backpedal Starting Stance

- Stand in a split stance, ensuring stability.
- Lower the center of mass by sinking the hips.
- Flex the front hip and knee to approximately 45 degrees.
- Flex the back knee to approximately 90 degrees and the hips to around 60 degrees.
- The front foot should be flat (firm base of support), while the rear foot heel should be raised. Distribute weight toward the balls of both feet.
- Keep the back flat.
- Lean the torso, with the shoulders over the front knee.
- The head is up looking forward (or toward the opponent; visual scanning).
- Relax the arms and hang them straight ahead of the hips.

First Step

- To initiate movement, drive off the back leg by extending the back leg knee and hip with the center of mass (COM) moving behind center of pressure at toe-off.
- Using a low heel recovery, the front leg moves back behind the rear leg and makes subsequent ground contact underneath the hips.
- Each foot contact should be toe to heel (never heel first).
- Keep hips low to increase stability.
- Maintain a forward trunk lean.
- Keep the head up and looking forward (or toward an opponent).
- Involve a relaxed contralateral arm movement to match the timing and amplitude of the leg movement. The range of motion at the shoulder should be small.
- The hips, knees, and ankles should be aligned in the frontal plane; avoid planting the foot wider than the hips to help maintain torso and pelvic position (i.e., hip step-width) and avoid a narrow step width.

Subsequent Steps

- With each step, use a low heel recovery to keep the feet close to the ground (i.e., rapid limb switching with minimal knee separation on ground contact) to enable a quicker transition to a subsequent movement (e.g., COD or forward sprint).
- Keep the hips low to maintain a low COM again to help quicken the transition into the subsequent movement (i.e., minimize vertical oscillation). Possible cues include “stay low” or “drop the hips.”
- Each foot contact should be toe to heel. A possible cue is “toe to heel.”
- Ensure foot contact is in line with the hips in the frontal plane to avoid feet crossing and the risk of trips or falls.
- Maintain a forward trunk lean.
- Keep the head up and looking forward (or toward an opponent). A possible cue is “eyes up” or “head up.”
- Use a relaxed, low-amplitude contralateral arm movement with the objective to project COM rearward, with minimal vertical projection and oscillation.

Side-Shuffle

Start Position

- Get in an athletic ready position: the feet are shoulder-width apart; weight is distributed toward the balls of the feet; flex the hips, knees, and ankles (i.e., strong quarter-squat position); keep the back flat with a torso lean (shoulders over knees); the head is up and the arms are by the sides with elbows flexed and hands in front of the body.

Foot Position

- Foot contact is on the balls of the feet.
- Feet are placed perpendicular to movement (parallel or slight hip external rotation).
- Feet stay close to the ground.
- Foot placement is wider than the knees; increased width increases stability and lowers COM.
- The feet should not cross. (In a diagonal or faster shuffle, the feet may cross using a crossover step to help with the directional change or provide extra propulsion to increase velocity [21].)

Leg Action

- Take short steps, driving more from the leg farthest (i.e., pushing the medial aspect of foot) from the intended direction of travel (i.e., COM moves medially relative to center of pressure at toe-off), with the other leg acting as a pivot, allowing the COM to travel medially and assist the shuffle step (COM is now over center of pressure).
- Leg action is close to the ground in preparation for ground contact.

COM Travel

- COM is projected medially or laterally, with minimal vertical projection and oscillation.
- The legs remain in a loaded athletic position and the trunk remains vertical.



Posture

- An athletic position is maintained throughout.
- Maintain flexion of the ankles, knees, and hips, lowering the COM.
- The back remains flat.
- The head is in line with the body.

Arm Action

- Perform a short, rapid arm action in the sagittal plane.

Training Goals

The three goals to enhance agility performance through training are to develop the physical qualities to allow athletes to adopt the required postures and positions for a range of different agility movements performed in the sport as well as enhance braking and propulsion of such tasks (action capacity); to develop movement efficiency and competency of range of agility movements pertinent to the athlete's sport, and to enhance perceptual-cognitive ability in various situations and tactical scenarios (35).

Methods of Developing Agility

As discussed throughout the chapter, a multifactorial approach is warranted to develop both the physical and the perceptual-cognitive aspects required for continual improvement in agility performance. Therefore, one should consider a plan that addresses underpinning physical capacities using various strength development strategies, technical competence in a closed-skill environment with movement-specific actions (COD ability), and improvement of perceptual-cognitive capacity through agility drills and inherent aspects of skills practice (58).

Strength

Similar to aspects of strength development for speed, strength development for agility should emphasize relative strength and a variety of speed–strength qualities along the force–velocity spectrum. However, additional development of the eccentric strength of the athlete due to the large braking forces during deceleration and COD actions within expressions of agility should be considered. This means that training exercises can include a spectrum of load–velocity profiles in the weight room (42) as well as a variety of load–velocity profile activities in the field, such as squat jumps, countermovement jumps, and drop jumps of various heights in combination with the COD and agility drills themselves. The multifactorial nature of agility warrants a well-rounded approach to strength development that spans the load– and force–velocity curve. [Table 20.4](#) provides an example of how training in the weight room and on the field can satisfy the strength requirements underlying agility performance. Understanding that athletes will develop the physical requirements before being able to transfer newly gained strength into technical competence helps coaches, strength and conditioning professionals, and athletes realize that performance gains will lag behind fundamental physical requirements for a given technical skill (99).

TABLE 20.4 Comparison of Focus for Agility Development in Novice and Advanced Athletes

Strength requirement	Novice athletes (weight room associated)	Novice athletes (field and court drills)	Advanced athletes (weight room associated)	Advanced athletes (field and court drills)
Dynamic strength: required to provide base strength for all subsequent training as well as to ensure adequate mobility during body weight and loaded training	Body weight exercises	Body awareness work such as leaning drills	• Squat (and variations) • Pulls (and variations) • Split-leg and unilateral variations	Various COD drills
Concentric explosive strength (may include isometric strength work): required to effectively reaccelerate (i.e., propulsion) after the braking phase or maintain strong position through the transition phase of COD and agility	Bilateral and unilateral box jumps	• Acceleration drills • Banded drills and isometric postural holds	• Box jumps • Weightlifting-based exercises • Squat jumps (loaded) • Lower-limb ballistic exercises initiated from static positions	Advanced acceleration drills (e.g., sled push)
Eccentric strength: required to develop the ability to effectively attenuate force, absorb energy, and support large external lower-limb flexion moments required during the	• Submaximal bilateral and unilateral landing stabilization exercises (i.e., drop landing, single-leg	Submaximal deceleration drills (forward emphasis and teaching various deceleration positions)	• Advanced bilateral and unilateral landing stabilization exercises performed in various planes • Drop landings and receiving	Maximal deceleration drills (high velocity and various angles)

Note: COD = change of direction.

Adapted by permission from S. Nimphius, "Increasing Agility," in *High-Performance Training for Sports*, edited by D. Joyce and D. Lewindon (Human Kinetics, 2014), 194.

Strength requirement	Novice athletes (weight room associated)	Novice athletes (field and court drills)	Advanced athletes (weight room associated)	Advanced athletes (field and court drills)
braking phase of COD and agility	hop, and hold) • Flywheel resistance training, lower-submaximal eccentrics, and tempo training		strength required during the catch phase of weightlifting-based exercises • Accentuated eccentric training • Maximal and supramaximal eccentrics and resisted eccentrics	
Reactive strength: required to increase the ability to transfer from high eccentric load to concentric explosiveness	Drop landing and landing stabilization exercises	Beginner plyometrics (slow to fast SSC activities, bilateral to unilateral, sagittal to multiplanar)	• Loaded jumps • Drop jumps • Complex or contrast training	Advanced plyometrics (fast SSC activities, unilateral, and multiplanar plyometrics)
Multidirectional strength: required to hold body position strongly during multitude of movement demands	Lunges, landmine skater squats	• Low-velocity COD (e.g., Z-drill) • Lateral, backward, and forward COD drills	• Unilateral exercises • Greater degree of freedom exercises (e.g., landmine exercises)	• High-velocity COD drills • Challenging cutting-angle COD drills
Perceptual-cognitive abilities: required	—	Simple reaction drills: introduction to	—	• Small-sided games

Note: COD = change of direction.

Adapted by permission from S. Nimphius, "Increasing Agility," in *High-Performance Training for Sports*, edited by D. Joyce and D. Lewindon (Human Kinetics, 2014), 194.

Strength requirement	Novice athletes (weight room associated)	Novice athletes (field and court drills)	Advanced athletes (weight room associated)	Advanced athletes (field and court drills)
to progress in visual scanning, effective anticipation, and decision making		increasing either temporal or spatial uncertainty		- Agility drills with severely restricted temporal and spatial aspects - Virtual reality and video-based training

Note: COD = change of direction.

Adapted by permission from S. Nimphius, "Increasing Agility," in *High-Performance Training for Sports*, edited by D. Joyce and D. Lewindon (Human Kinetics, 2014), 194.

Change-of-Direction Ability

Similar to the progressions of plyometric activities based on intensity and difficulty of each drill, closed-skill COD drills can progress from beginner to intermediate to advanced levels based on the physical loading demands that have been discussed in this chapter (regulation of angle and intensity, which affect mechanical loading) (36). [Table 20.5](#) presents an example of a progression of drills for COD ability (both rapid COD and those associated with maneuverability), as well as a progression plan for agility drills. The physical loading during agility drills will be higher than beginner COD drills. Therefore, [table 20.5](#) provides practitioners with a series of drills and a guideline for safely and effectively advancing athletes so they can build on their physical and technical attributes as they improve.

TABLE 20.5 **Change-of-Direction and Agility Drill Progressions**

Agility component	Beginner	Intermediate	Advanced
COD	• Deceleration drills (forward) progressing to higher entry velocity or shorter distance to stop • Basic movement patterns for forward, backward, and lateral shuffling • COD drills that involve low velocities (less than 5 yd [5 m] of acceleration into COD)—Z-drill, for example	• Deceleration drills (lateral) with same progressions as forward • Expand to include a broad range of cutting angles less than 75 degrees • May increase entry velocity during drills (up to 10 yd [9 m] of acceleration leading into COD)	• Deceleration to reacceleration in both forward and lateral directions • Expand further to a comprehensive range of cutting angles, including those greater than 75 degrees • Maximal-effort decelerations and entry velocities
Maneuverability	Basic drills or tests such as the Illinois agility that require nearly straight-line running with slight bends	• Drills that increase the difficulty of the bend involved such as the L-run • Drills that require transition between modes of movement (shuffling, sprinting, and backpedaling) such as the T-test	Repeated maneuverability activities with manipulated work-to-rest ratios
Agility	Physical and technical competence should occur before agility drills are incorporated	• COD drills in beginner and intermediate categories with the addition of simple stimuli (arrow, pointing in a certain direction)	• Expand into large degrees of spatial and temporal uncertainty (and therefore greater perceptual-cognitive stress)

Note: COD = change of direction.

Adapted by permission from S. Nimphius, "Increasing Agility," in *High-Performance Training for Sports*, edited by D. Joyce and D. Lewindon (Human Kinetics, 2014), 194.

Agility component	Beginner	Intermediate	Advanced
		These drills present a limited number of options for the athlete to have to react to (e.g., right or left, forward or back) on the signal and do not develop agility per se; however, they further increase intensity of drills along the agility development continuum.	- Small-sided games - Evasion games and drills

Note: COD = change of direction.

Adapted by permission from S. Nimphius, "Increasing Agility," in *High-Performance Training for Sports*, edited by D. Joyce and D. Lewindon (Human Kinetics, 2014), 194.

Perceptual-Cognitive Ability

During years of skills training (within the sport), athletes continually enhance visual scanning, pattern recognition, and knowledge of the situation, while in drills used to improve agility (outside of skills practice) they primarily focus on improving anticipation, decision-making time, and accuracy. In the same way one would regularly increase the demands of a physical task to continually improve performance, one can train perceptual-cognitive ability. Therefore, agility activities should begin by adding a **perceptual-cognitive** component (i.e., progressing from low to high cognitive loading) to common closed-skill COD drills. For example, decelerations or the Z-drill can evolve into agility drills by including a generic stimulus such as a whistle, a voice command, or a flashing arrow or light (87, 110). Following success with the aforementioned agility drills, progression requires activities that have sport-specific stimuli such as evasive drills and small-sided games, which have been said to have better transfer to performance (87, 110). Both generic and specific stimuli can be made more difficult within a given agility drill through progressive increases in either temporal (time) or spatial (space or

area) stress on the athlete. See [table 20.6](#) for a range of agility factors that can be manipulated in agility drill design.

TABLE 20.6 Mechanical and Perceptual-Cognitive Factors That Can Be Manipulated During Agility Drills

Variable	Factor	Interpretation
Mechanical	Spacing and pitch or court dimensions	Altering the space and pitch or court dimensions will vary the distances traveled and subsequent entry and exit velocities and momentum achieved, and therefore allow athletes to develop agility from lower velocity, intensive scenarios, to higher velocity, extensive scenarios. Additionally, entry velocity regulates intensity and subsequent mechanical loading experienced.
	Entry and exit velocity momentum	
	Agility actions and angle of COD	Practitioners can put restrictions on the type of COD action performed or allow athletes to try a range of different COD actions to develop between-skill variability and movement solutions. Additionally, varying the location of end or target zones can allow athletes to develop shallow, moderate, and sharp COD ability to enable them to be proficient in changing direction across a spectrum of angles.
	Starting position (static or transitional movements)	Athletes can begin the drill from different static initiation (e.g., drop step, jab step) and from different transitional movements (i.e., backpedal, crossover running, lateral shuffle) to create drill variability and to increase game representativeness and ability to evade or defend from different locomotor actions.
Perceptual-cognitive	Location and number of attackers or defenders	The location of the defenders and attackers can be manipulated. For example, in addition to front-on situations, defenders can be located to the side or behind the attackers. This is pertinent for 360-degree sports in which defenders and attackers can be located in various locations on the pitch or court. The addition of multiple attackers and defenders can increase game representativeness, allowing the opportunity for decoy runs; attacking and defensive transitions and set plays; team communication; and increased chaos, task complexity, cognitive loading, and variability.

Note: COD = change of direction.

Variable	Factor	Interpretation
	Temporal constraints	Temporal or time constraints can be administered (i.e., athlete has 5 s to achieve the objective) to maintain quality and intensity (and often to replicate the reduced temporal constraints associated in random sports). Work-to-rest ratios can be manipulated for energy system development and to represent peak demands of sports.
	Rules and conditions	Practitioners can be creative and vary the rules (i.e., one-hand or two-hand tags) and put constraints on the attackers or defenders to create variation (i.e., only two evasive actions can be performed before committing to a specific direction within the drill).
	Environment and surfaces	To increase game representativeness, drills should be performed on the sport-specific surface. However, certain sports may play on different surfaces (i.e., grass and artificial turf) with different hardness and frictional properties, and thus these should be considered.
	Generic to sport-specific stimulus (developing representative learning environments)	Generic stimuli and sources of information can be used such as auditory commands, generic visual commands, and stimuli such as flashing lights, arrows, and coach- or player-dictated directional commands to increase cognitive loading and increase mechanical loading in a controlled environment. Typically, the timing of stimuli can be controlled as well using specific equipment. However, to increase game representativeness, develop perceptual-cognitive speed, knowledge of situations, memory recall, and anticipation, sport-specific stimuli (i.e., multiple attackers or defenders, inclusion of sport-specific implements and scenarios) can be included to develop movement solutions and transfer to sporting performance. These drills can be developed collaboratively with the skills coach for simultaneous technical and tactical development.
	Task complexity and cognitive loading	To challenge more advanced athletes and to ensure athletes are prepared for the chaotic demands of sport when returning to sports, drills can begin with relatively simple closed and open drills with one or two stimuli and sources of information, before progressing to more chaotic open drills of increased game representativeness with various sources of stimuli and information (i.e.,

Note: COD = change of direction.

Variable	Factor	Interpretation
		multiple attackers or defenders, inclusion of sport-specific implements and stimuli can be included).

Note: COD = change of direction.

Program Design

Program design involves planning at several levels, each with its own set of considerations: microcycles (short term), mesocycles (medium term), and macrocycles (long term). The process is referred to as *periodization*. **Periodization** is the strategic manipulation of an athlete's preparedness through the employment of sequenced training phases defined by cycles and stages of workload (see [chapter 22](#)). These planned workloads are varied to facilitate the integration of planned programming tactics that will harmonize the relationship between training-induced fatigue and accommodation (30). Furthermore, the periodization process is guided by information collected during the athlete monitoring process. Specifically, as an athlete progresses through a training phase, consistent information acquired through various monitoring strategies helps to show how the athlete is responding to the training stimulus in the short term and how the athlete should be further developed in the long term—that is, how the original training program should be modified to reflect responses to the training.

At each level of training plan development, strength and conditioning professionals must manipulate certain variables. While these are useful in characterizing the training agenda as well as quantifying workloads, strength and conditioning professionals should keep in mind that each variable must be considered individually, and they should regard how it affects the collective system. Moreover, individual athlete training responses, which athlete monitoring reveals, should be used in the manipulation of these variables:

- *Angle*—The intended angle for COM deflection during the task and a regulator for knee joint loading
- *Approach velocity*—The horizontal entry velocity of the COM as the athlete begins to initiate their COD action over multiple steps and a regulator for knee joint loading

- *Exercise (or work) interval*—The duration or distance over which a repetition is executed
- *Exercise order*—The sequence in which a set of repetitions is executed
- *Frequency*—The number of training sessions performed in a given time period (e.g., day or week)
- *Intensity*—The effort with which a repetition is executed (% of maximum)
- *Recovery (rest) interval*—The time period between repetitions and sets
- *Repetition*—The execution of a specific workload assignment or movement technique
- *Series*—A group of sets and recovery intervals
- *Set*—A group of repetitions and rest intervals
- *Volume-load*—The density of volume performed at prescribed intensities—for example, 3 sets of 5 repetitions at 100 kilograms results in a volume-load of 1,500 kilograms
- *Volume*—The amount of work (e.g., 3 sets of 5 repetitions) performed in a given training session or time period
- *Work-to-rest ratio*—The relative density of exercise and relief intervals in a set, expressed as a ratio

[Chapter 22](#) explains periodization of resistance training programs, and the same concepts can be applied to sprint, COD, and agility training.

Speed Development Strategies

The planning process for speed development is similar to that used by strength and conditioning professionals in the weight room. Specifically, planning tactics should be periodized in a manner that addresses the physical and psychological components of sprinting through emphasis and de-emphasis on particular qualities in a phasic manner. An athlete's capability to sprint can be improved through the incorporation of training periods that are designed to fully maximize and saturate a fitness quality, which may bolster the effects of future training agendas.

To enhance speed, a practitioner should consider the relationship between human locomotion and force production. Throughout this chapter the authors have pointed to a link between sprint success and certain physiological phenomena. Elite sprinters produce large amounts of force within a short amount of time. This high level of force allows the sprinter to produce longer step lengths at a faster rate. This display of movement on the track reflects enhanced neuromuscular factors—namely, maximal strength, RFD, and impulse. These variables are the product of well-designed training protocols that aim to enhance qualities such as the hypertrophy of task-specific motor units, firing frequency, rate coding, and muscle–tendon stiffness.

An option for speed development is provided in [table 20.7](#), demonstrating how sequenced phases of training may seamlessly develop sprint potential through the maturation of accelerative abilities before the execution of maximal-velocity efforts. This model, which can be referred to as a *short to long method* of sprint training, is a conceptual progression that attempts to merge the relationship between larger rates of acceleration and greater sprint velocities (discussed earlier in this chapter) (31). Specifically, an athlete taking part in a short to long model will begin the training year with an emphasis on improving propulsive force output through short sprints that maintain the biomechanics associated with the acceleration

phase of a sprint. Later, the athlete bridges into longer sprint work that aims to enhance top speed through upright running mechanics. The concept behind this proposed model is that an athlete may see greater improvements in top speed if force production can occur at the ideal rate and moment through an investment in acceleration development and optimization (62). An alternative approach for track sprinters (100-200 m) may involve sequencing training from general to specific preparation starting with an aerobic foundation to developing anaerobic capacity (short speed–endurance), to speed–endurance (long speed–endurance) culminating in speed (inclusive of acceleration and maximum velocity) before the main competition (14, 46).

TABLE 20.7 Application of Speed Development Strategies: Phasic Development for Short-Distance Sprinting (e.g., 100 m sprinter; bobsled athlete)

Block and emphasis	Training tool and notes	Load prescription (rest periods between sets) for an example microcycle within each block ^a
Block 1 Primary aim: Acceleration development	1. Use incline sprinting to develop proper acceleration mechanics, providing an opportunity to maximize propulsion forces in the initial stages of a sprint. 2. Emphasize low heel recovery; driving the foot down and through the ground; aggressive and complete arm action; neutral head position in line with the spine. Resisted acceleration bounds could be incorporated to emphasize lower limb extension and forceful thigh retraction during late swing.	Monday: Incline sprints 1 × 3 × 15 m (1.5 min) 1 × 4 × 20 m (2 min) Wednesday: Resisted or sled acceleration bounds (2 min) 1 × 4 × 15 m 1 × 2 × 20 m (2 min) Friday: Incline sprints 1 × 2 × 15 m (1.5 min) 1 × 3 × 20 m (2 min)

Block and emphasis	Training tool and notes	Load prescription (rest periods between sets) for an example microcycle within each block ^a
		1 × 2 × 25 m (2.5 min)
Block 2 Primary aim: Long acceleration development Secondary aim: Improve transition to upright sprinting	1. Introduce flat-ground sprinting from a low starting position or towing mechanism. 2. Promote the correct biomechanics of transition sprinting from acceleration to maximal velocity through the inclusion of acceleration holds. ^b	Monday: Incline sprints 1 × 3 × 30 m (3 min) 1 × 3 × 40 m (4 min) Wednesday: Push-up starts 1 × 4 × 15 m (2 min) Sled tows 2 × 3 × 20 m (2 min /4 min) Friday: Crouch starts 1 × 4 × 15 m (2 min) Acceleration holds 1 × 3 × 40 m (4 min)
Block 3 Primary aim: Introduction to max speed training Secondary aim: Maintain accelerative ability Tertiary aim: Exposure to speed–endurance; introduction to special endurance	1. Maintain acceleration abilities through inclusion of short sprints at the beginning of most sessions. 2. Begin exposure to maximum speed training through exercises that use upright sprinting mechanics. 3. Prescribe speed–endurance or special endurance sessions later in the week when effects of summated fatigue may dampen neural drive.	Monday: Block starts 1 × 2 × 20 m (2.5 min) 1 × 2 × 30 m (3.5 min) Accelerations from crouch stance 1 × 3 × 40 m (5 min) Wednesday: High-stance starts 1 × 3 × 20 m (2.5 min) Fly-ins 1 × 4 × 20 m build/20 m fly (4 min) Friday: Block starts 1 × 4 × 20 m (3 min)

Block and emphasis	Training tool and notes	Load prescription (rest periods between sets) for an example microcycle within each block ^a
		Accelerations from crouch stance 1 × 2 × 40 m (5 min) Speed–endurance from two-point stance 1 × 2 × 60 m (5.5 min)
Block 4 Primary aim: Enhance maximal speed ability Secondary aim: Maintain accelerative ability Tertiary aim: Continued exposure to speed–endurance and special endurance	1. Continue exposure to maximum speed training through longer fly-in zones or race-modeling practices. 2. Maintain acceleration abilities through incorporation of short sprints at the beginning of most sessions. 3. Prescribe speed–endurance or special endurance sessions relevant to the athlete’s primary events.	Monday: Block starts 1 × 2 × 30 m (3.5 min) Acceleration holds 1 × 2 × 40 m (5 min) Fly-float-fly 1 × 4 × 20/20/20 (7 min) Wednesday (technical session): Ankling 2 × 15 m (1 min) A-skip 2 × 20 m (1 min) Ankle dribble 1 × 20 m (1 min) Calf dribble 1 × 20 m (1 min) Knee dribble 1 × 20 m (1 min) Dribble bleed 1 × 30 m Straight-leg bound 2 × 20 m (2 min) Wicket ^c 2 × 30 m (3 min) Wicket with run-off 2 × 30 m + 15 m (4 min) Dribble blend 2 × 2 × 40 m (5 min)

Block and emphasis	Training tool and notes	Load prescription (rest periods between sets) for an example microcycle within each block ^a
		Friday: Block starts 1 × 3 × 25 m (3 min) Speed–endurance from two-point stance 1 × 2 × 50 m (6 min) 1 × 2 × 70 m (6.5 min) 1 × 2 × 90 m (7 min)

^aEach session should be preceded by a warm-up that incorporates a range of technical drills to work on aspects of technique in relation to both acceleration and maximum-velocity sprinting (see sprint drills 1 to 6 on [pages 675-683](#) for examples). These drills could also form dedicated technique sessions as depicted in block 4 of the table.

^bAn acceleration hold is a sprint drill for which the strength and conditioning professional places a cone on the track near or slightly before the point at which an athlete's shin typically rises to a vertical position, which is indicative of upright sprinting. The athlete is then instructed to maintain the speed achieved up to the cone through the remainder of the sprint. For instance, an athlete may be prescribed acceleration holds for 40 meters, but the coach's mark ending the acceleration zone is placed at the 20-meter mark on the track. The athlete then maintains that speed from 20 meters to 40 meters. This drill is used to begin to focus on improving transitional mechanics.

^cWickets are a constrained sprinting task that involves a period of sprint running whereby the athlete appears to step over objects (e.g., mini-hurdles) in an effort to encourage high foot recovery and optimal front-side mechanics.

Biomechanical Differences in Acceleration Between Sprint and Team-Sport Athletes

The ability to accelerate linearly and displace the COM rapidly is essential in track sprinting and multidirectional team sports. Track sprinters are the fastest humans on earth, and thus examining the biomechanics and techniques they adopt is intuitive to better understand and optimize sprinting performance in athletes. Clear biomechanical and technical differences exist between sprint and team-sport athletes (25, 107); however, it is important to consider that environment, task, and performer constraints will all interact and influence the acceleration strategy and mechanics employed and the overall execution of the sprinting skill.

Wild and colleagues (107) examined the first three steps during initial acceleration and observed that sprinters achieved greater step velocities and greater normalized average horizontal external power, and they displayed greater normalized step length and flight length, greater normalized touchdown distance (i.e., foot impact under or behind COM), and greater toe-off distance (i.e., effective push-off behind COM) compared to rugby forwards and backs. Conversely, rugby backs displayed slightly shorter contact times and slightly greater step rates compared to sprinters. From a kinetics perspective for later acceleration at matched velocities (step 8-12 for sprinters and 9-20 for soccer), while no differences were observed in the initial 20% of the stance phase between sprinters and soccer athletes, sprinters produced greater anterior-posterior force across most of the propulsive phase and greater net forces over similar contact times, thus resulting in greater relative net impulse (25). As such, the sprinters achieved greater average horizontal external power, which allowed them to accelerate beyond the velocity plateau of the soccer athletes.

It should be noted that the spatiotemporal, kinematic, and kinetic differences observed could be partially attributed to sprinters accelerating from a four-point position using blocks and performed on a rubber surface while wearing spikes. This results in different body segment orientations such as greater forward trunk lean and horizontal shin angles, which contribute to more horizontally orientated force (107). Conversely, team-sport athletes initiate their sprints from a more upright, two-point staggered-stance position (resulting in a more vertically orientated force), on a different surface with less favorable frictional properties for achieving peak speed (i.e., grass, field turf, indoor courts), and with different footwear that is designed for multidirectional and sport-specific movements (e.g., cleats or trainers). Additionally, sprinters' preparation strategies are designed around the sole task of maximizing linear speed, which is the aim of that sport, and an isolated event. As such, sprinting is a skill, and thus, sprinters can dedicate more time and deliberate practice to developing the skill of sprinting, while their periodized strength and conditioning programs can also be tailored to maximize the physical qualities that underpin linear sprinting. Conversely, team-sport athletes must be proficient at a range of multidirectional movements (i.e., changing direction, curved sprints of varying radii and direction, compared to sprinters who perform curved sprints anticlockwise with radii ≥ 40 yd [36.5 m], and deceleration) and sport-specific skills (e.g., passing, tackling, shooting). Additionally, team-sport athletes must perform technical and tactical training and generally must do so repeatedly over the course of long matches (generally ranging 40-90 mins) and competitive seasons. Due to these competing demands, team-sport athletes must spend a significantly lower proportion of time developing sprinting skills, while their preparation programs also differ due to the different physical, biomechanical, and cardiorespiratory fitness demands between sports.

Monitoring Sprint Ability

The longstanding method of assessing an athlete's speed ability has typically involved a test of maximal-effort linear sprinting. Much of the time, the maximal-effort sprint test is carried out over a distance, such as 40 yards (37 m), that is believed to be relevant to the athlete's sport. While time to completion is an effective tool to evaluate changes in sprint performance, these tests do not fully address the minute changes regarding an athlete's progression toward race or field readiness. In addition, testing error in the form of hand timing and various start tactics confound the interpretation of testing results. As a result, strength and conditioning professionals may consider the adoption of additional monitoring tools as part of their battery of assessments.

One such method of evaluation comes from the assistance of high-speed cameras or optical timing systems that use interruptions of infrared light to capture data from an athlete's collective foot strike (15). This technology provides strength and conditioning professionals with additional insight into the sprint capabilities of the athlete, such as technique and movement strategies (16). Additionally, technology that permits the quantification of instantaneous speed is also advantageous for insights into speed at specific distances, time to achieve specific speeds, peak speed, acceleration, and τ (acceleration time constant). [Table 20.8](#) provides a description of key variables that may be beneficial in the monitoring of speed development.

TABLE 20.8 Monitoring Speed Development

Measure	Description	Interpretation
Ground contact time	The total time allotted for a single stance phase.	In general, decreased ground contact times coupled with improved sprint completion times may suggest that an athlete has improved the ability to generate high rates of force in a shorter

Note: ETG = electronic timing gate; 2D = two-dimensional; 3D = three-dimensional; LiDAR = light detection and ranging.

Measure	Description	Interpretation
		amount of time. However, an athlete may demonstrate slightly longer ground contact times at the initial stages of the sprint due to acceleration mechanics.
Step length	The distance between the toe and heel of two consecutive footsteps. Example: centimeters between the heel of the right foot and the heel of the left foot.	Improvements in step length (increases in distance covered) may occur through improved sprint mechanics or posture and ultimately enhanced force output.
Stride length	The distance between the heel in two consecutive steps of the same foot. Example: meters between the heel of the right foot and the heel of the next ground contact made by the right foot.	Improvements in stride length (increases in distance covered by two successive steps) may occur through improved sprint mechanics or posture, and ultimately enhanced force output.
Flight time	The duration of time that an athlete is not in contact with the ground. This occurs between the end of one step and the beginning of the subsequent step.	Increases in flight time can be interpreted in two ways: • An athlete is producing higher vertical forces over a shorter amount of time, resulting in optimal horizontal displacement of the athlete's mass down the track. • An athlete is producing higher forces over a longer amount of time, which may result in unnecessary vertical amplitude of the center of mass.
Stride angle	The angle at which the foot leaves the track.	The angle of stride may provide the strength and conditioning professional with the ability to determine if the athlete is producing forces at the right moment, which is achieved through proper sprint mechanics. Furthermore, data on the stride angle elucidate what stage of sprint the athlete is attaining (acceleratory, transitory, maximum speed) and may relate to ground contact times.

Note: ETG = electronic timing gate; 2D = two-dimensional; 3D = three-dimensional; LiDAR = light detection and ranging.